Shallow Probes Are Sufficient for Near-Oracle Personalization Depth Allocation in Federated Learning

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***Abstract*—Most personalized federated learning (PFL) work asks how to personalize a shared model. Rather than proposing a new federated optimizer, we study whether a small client-specific measurement budget is sufficient to support cost-aware personalization-depth decisions under a utility that trades Macro-F1 against deployment cost. Exhaustive per-client evaluation of six personalization depths defines an offline oracle, an empirical upper bound that is not deployable. We introduce and empirically audit the Probe Sufficiency Hypothesis: three shallow probes at the global, head-only, and shallow-layer depths may contain most of the decision-relevant information required to approach that oracle. Our primary contribution is the empirical formulation and audit of probe-first personalization-depth allocation; FedPBA (Federated Probe-Based Allocation) is a learned instantiation used to test whether shallow probe evidence supports extrapolation to unmeasured deeper actions. On CIFAR-10, CIFAR-100, and a patient-disjoint PTB-XL ECG subset (45 federated sessions; 900 clients; 2700 client-cost evaluations; session-grouped GroupKFold), a training-free greedy probe rule achieves mean regret 0.0071 and recovers 98.3 percent of oracle utility, outperforming both metadata-only policies and the more complex learned allocator (0.0110 / 97.3 percent). Best uniform and best context-only reach only 0.0483 / 88.4 percent and 0.0476 / 88.5 percent. Aggregate shallow-best versus oracle correlation is high (0.9818) but partly structural (2567 of 2700 shallow-oracle rows); on deep-oracle rows alone, the correlation is 0.578. Deep-oracle PTB-XL cases remain a high-regret boundary, and no validated escalation detector is provided.**

***Index Terms*—federated learning, personalization-depth allocation, probe sufficiency, probe-first allocation, deployment cost, regret, resource-aware FL**

I. Introduction

Federated learning (FL) trains models across decentralized clients without pooling raw data [1], [2]. Personalized FL (PFL) then adapts the shared model to heterogeneous clients via meta-learning, proximal objectives, personalization layers, or representation sharing [3]–[6]. That literature primarily addresses three related but distinct questions only partially:

(A) *How* should a shared model be personalized (objectives, layers, adapters, regularization)?
(B) *How deeply* should a particular client personalize under an explicit performance–cost utility?
(C) Can that depth be selected *without* exhaustively evaluating every candidate depth?

Most PFL work addresses Question (A). This paper focuses on Questions (B) and (C): cost-aware per-client personalization-depth allocation as a distinct decision problem, separate from the design of a new personalized federated optimizer.

Deeper local adaptation can raise Macro-F1 but increases trainable parameters, FLOPs, wall time, communication, and memory. Exhaustively evaluating all personalization depths per client (the offline *oracle*) is informative yet not deployable: it requires six depth evaluations (one global-model evaluation at P_0 plus five personalization train-and-evaluate runs at P_1–P_5) before any depth is chosen. A single uniform depth is inexpensive but ignores client-specific accuracy–cost curves.

A companion characterization study [23] primarily studied the offline decision landscape: when personalization depth is worth its cost relative to an exhaustive oracle. The present paper goes beyond that characterization. Its scientific object is the information-sufficiency and policy-design problem required to *approximate* that landscape under a constrained measurement budget: probe-derived information, policy-class comparison, deep-action extrapolation, failure conditions, and deployable approximation of the offline oracle.

A natural alternative is to measure only the early portion of each client's local utility curve. After FedAvg, probes at P_0–P_2 observe how a client responds to no adaptation, head adaptation, and limited representation adaptation. We introduce the Probe Sufficiency Hypothesis: a small set of client-specific shallow utility measurements may contain most of the information required to approach the exhaustive offline oracle. Probe-first allocation separates measurement from commitment: a small local measurement budget is spent before selecting a potentially expensive personalization action. This resembles budgeted computation only as a systems principle and is not a novelty claim over early-exit literature.

This claim is not trivial. (i) Deep personalization can yield large accuracy gains on some clients. (ii) Gains from P_3–P_5 are not guaranteed to be predictable from P_0–P_2. (iii) Dataset identity, Dirichlet α, and λ alone may not separate clients within

the same federation. (iv) The oracle's preference for shallow depths is observable only after all six depths are evaluated; it is unknown at deployment time.

We therefore audit four policy classes under

$$U_i(a) = \text{Macro-F1}_i(a) - \lambda \, \text{Cost}_i(a). \tag{1}$$

Fixed policies use no client-specific probe information. Context-only learners map metadata $\mathbf{c}_i$ to depth without measuring $\mathbf{s}_i = [U_i(P_0), U_i(P_1), U_i(P_2)]$. The training-free greedy probe rule is the direct sufficiency test: after three measurements, deploy $\arg\max_{a \in \{P_0, P_1, P_2\}} U_i(a)$. FEDPBA (Federated Probe-Based Allocation) is a complementary learned probe allocator that uses the same probes plus context to test deep-action extrapolation to P_3–P_5 without evaluating those depths. We do not present greedy and FEDPBA as co-equal proposed algorithms: greedy is the primary empirical test of Probe Sufficiency, and FEDPBA is a learned instantiation of the probe-first framework.

Contributions.

1) **Personalization-depth allocation problem.** We formulate cost-aware, per-client personalization-depth allocation as the problem of approximating an exhaustive offline oracle using fewer local evaluations. This is distinct from proposing a new PFL optimizer.
2) **Probe Sufficiency Hypothesis.** We introduce an empirical hypothesis that utilities measured at P_0, P_1, and P_2 may contain most of the decision-relevant information required to approach the oracle.
3) **Probe-first policy audit.** We evaluate fixed, context-only, training-free greedy-probe, and learned probe-based policies under a common utility, grouped evaluation protocol, and session-level inference, isolating whether gains arise from metadata, client-specific measurement, or allocator complexity.
4) **Learned instantiation.** We instantiate a learned probe allocator, FEDPBA, using shallow probe outcomes plus context to predict among P_0–P_5. The tabular learner is an implementation of the framework rather than the source of algorithmic novelty.
5) **Evidence and boundary conditions.** In the evaluated setting, greedy probe achieves mean regret 0.0071 and 98.3% oracle utility recovery; FEDPBA reaches 0.0110 / 97.3%; aggregate shallow-best vs. oracle correlation is r=0.9818 (deep-oracle-only r=0.578). PTB-XL and deep-oracle cases remain the main failure boundary; no validated escalation gate is provided.

This paper proposes neither a new federated optimizer nor a new cryptographic privacy mechanism [25]. Any privacy benefit inherits only from FL's raw-data locality.

II. Related Work

A. Personalized Federated Learning

FedAvg and heterogeneity-robust variants learn shared models under data locality [1], [10], [11]. PFL methods design objectives, representations, proximal adaptation, meta-learning, or personalized layers [3]–[8]. **Gap:** the missing question is not how to construct a personalized model, but whether a small client-specific measurement budget can support near-oracle selection among personalization depths under an explicit performance–cost utility.

B. Resource-Aware Federated Learning

Client selection, compression, heterogeneous capacity, scheduling, and hardware-aware federated deployments reduce federation-level cost [12]–[16], [24]. **Gap:** these address *who* participates or *how* to communicate under resource constraints. After global training, we ask *how deeply* each participating client should adapt under a cost-aware utility, and whether that depth can be chosen with a limited probe budget.

C. Adaptive and Parameter-Efficient Personalization

Recent PFL and federated PEFT work personalizes *how* adapters are structured or aggregated rather than whether a shallow measurement budget can approximate an offline depth oracle. FedDPA couples global and local adapters with instance-wise weighting for test-time personalization [17]. FlexLoRA aggregates heterogeneous LoRA ranks under client resource constraints [18]. FedTT reduces communication via tensorized adapters [19]. FedP^{2}EFT learns client-specific PEFT structures (e.g., LoRA ranks) collaboratively [20]. Complementary dual-/adaptive-LoRA designs further personalize federated fine-tuning under heterogeneity [21], [22]. **Gap:** to the best of our knowledge, the evaluated literature does not treat three shallow client-specific utility measurements as the central control signal for approximating a cost-aware personalization-depth oracle.

D. Budgeted Computation Perspective

Probe-first allocation resembles early-exit / budgeted computation as a systems principle: spend a small measurement budget before committing to a costly action. Here the stopping decision is made at the client-level personalization-depth allocation layer, not at an inference-layer exit. We use this analogy only as conceptual framing, not as a novelty claim over early-exit methods.

E. Positioning Relative to Prior Characterization

A companion characterization study [23] defined and characterized the offline decision problem: the utility gap between fixed-depth policies and a per-client exhaustive oracle. The present paper investigates an operational approximation of that oracle using restricted client-specific shallow measurements. The experimental stack is shared where noted, but the scientific object is new: probe-derived information, policy-class comparison, deep-action extrapolation, failure-condition analysis, and deployable approximation of the offline oracle, rather than another fixed-depth characterization.

III. Problem Formulation

We distinguish *decision formulation* from *decision policy*. Decision formulation defines the action space, utility, cost model, and offline oracle. Decision policy selects an action using information available before exhaustive evaluation. The companion characterization study primarily studied the former;

TABLE I
PERSONALIZATION LADDER (ACTION SPACE).

Action	Locally trainable components
P_0	None (global-only)
P_1	Classifier head
P_2	Adapter and head
P_3	Last encoder block, adapter, head
P_4	Last two encoder blocks, adapter, head
P_5	Full model

the present paper studies the latter under a constrained shallow probe budget.

A. Personalization Ladder and Cost-Aware Utility

Clients $i \in \{1, \ldots, N\}$ obtain a global model by FedAvg, then select $a \in \mathcal{A} = \{P_0, \ldots, P_5\}$ (Table I). Utility follows Eq. (1). Macro-F1 is used because client-level label imbalance is common under Dirichlet partitions and in medical classification; we did not compare Accuracy or AUROC in this study.

Cost model. For each action we record trainable parameters, FLOPs, local training wall time, communication bytes, and a memory proxy. Raw cost is a weighted sum of magnitude-scaled components,

$$\begin{aligned}\mathrm{Raw}_i(a) = {} & w_{\mathrm{param}} C_{\mathrm{param},i}(a)/10^5 + w_{\mathrm{flop}} C_{\mathrm{flop},i}(a)/10^8 \\ & + w_{\mathrm{time}} C_{\mathrm{time},i}(a) + w_{\mathrm{comm}} C_{\mathrm{comm},i}(a)/10^6 \\ & + w_{\mathrm{mem}} C_{\mathrm{mem},i}(a)/100,\end{aligned} \tag{2}$$

with default weights $(w_{\mathrm{param}}, w_{\mathrm{flop}}, w_{\mathrm{time}}, w_{\mathrm{comm}}, w_{\mathrm{mem}}) = (0.25, 0.20, 0.25, 0.15, 0.15)$, then multiplied by a client hardware multiplier. Costs are min–max normalized *within each client across the six actions*:

$$\widehat{\mathrm{Cost}}_i(a) = \frac{\mathrm{Raw}_i(a) - \min_b \mathrm{Raw}_i(b)}{\max_b \mathrm{Raw}_i(b) - \min_b \mathrm{Raw}_i(b)}, \tag{3}$$

with the convention that if the range is below 10^{-12}, all six normalized costs are set to 0. Min–max serves to combine heterogeneous physical units into one utility; λ then sets the exchange rate between Macro-F1 and normalized cost. Client-wide scalar hardware multipliers cancel under Eq. (3); the present cost model is therefore *not* hardware-aware for action ranking, and action-dependent hardware profiling remains future work. We treat $\lambda \in \{0.1, 0.3, 1.0\}$ as evaluation regimes, not universal physical constants. A post-hoc sensitivity analysis over six weight profiles and two normalizations (aggregate-first N1; component-first N2) yields greedy regret in $[0.0070, 0.0094]$ and recovery in $[97.7\%, 98.3\%]$ with shallow-oracle mass in $[91.3\%, 95.3\%]$; under these tested profiles, the aggregate probe-sufficiency result remains stable.

The offline oracle $a_i^\star = \arg\max_{a \in \mathcal{A}} U_i(a)$ requires all six depths and is an empirical upper bound only. Regret is $r_i(\pi) = U_i(a_i^\star) - U_i(\pi_i)$. Mean oracle utility over the evaluation set is positive (0.4147), so the recovery ratio below is well-defined; recovery would be fragile if the oracle aggregate approached zero or became negative.

B. Probe Sufficiency Hypothesis

Define the probe vector $\mathbf{s}_i = [U_i(P_0), U_i(P_1), U_i(P_2)]$ and context $\mathbf{c}_i$ (dataset, Dirichlet α, λ). These probes are meaningful because P_0 measures the unadapted global model, P_1 measures head-only adaptation, and P_2 measures limited representation adaptation; together they sample the early shape of the local utility curve.

Probe Sufficiency Hypothesis. For allocation under Eq. (1) in the evaluated setting, $\mathbf{s}_i$ contains most decision-relevant information for approaching the oracle. Sufficiency is empirical, not theoretical.

We operationalize "most decision-relevant information" as: (i) lower regret than context-only and fixed policies; (ii) high oracle utility recovery

$$\mathrm{Recovery}(\pi) = \frac{\sum_i U_i(\pi_i)}{\sum_i U_i(a_i^\star)}; \tag{4}$$

(iii) strong association between shallow-best utility and oracle utility. These are empirical criteria, not a theorem.

Three related but distinct notions should not be conflated: **aggregate value agreement** ($\max(U_i(P_0), U_i(P_1), U_i(P_2))$ tracks $U_i(a_i^\star)$, r=0.9818); **shallow-action frequency** ($a_i^\star \in \{P_0, P_1, P_2\}$ in 2,567/2,700 cases); and **deep-regime predictability** (whether shallow signals predict deep gains when $a_i^\star \in \{P_3, P_4, P_5\}$). These are not interchangeable evidence. Aggregate r=0.9818 is partly structural because shallow-best equals oracle utility whenever the oracle is shallow. *High aggregate correlation is not equivalent to validated prediction of deep personalization gains.*

C. Policy Classes Audited

(A) Fixed policies: global-only (P_0); personalize-all (P_5); majority (mode of oracle actions on the training partition); best uniform (one depth per configuration). No client-specific probe information.

(B) Context-only: learners mapping $\mathbf{c}_i \to a$ without $\mathbf{s}_i$ (metadata but no measured client-specific utility).

(C) Training-free greedy probe: evaluate P_0–P_2; deploy $\arg\max_{a \in \{P_0, P_1, P_2\}} U_i(a)$. This is the direct sufficiency test of the Probe Sufficiency Hypothesis.

(D) Learned probe allocator (FEDPBA): a meta-level allocation classifier on probe outcomes and context that may predict P_3–P_5, testing whether measured shallow evidence supports deep-action extrapolation. FEDPBA is not a new federated optimizer and does not alter FedAvg training; it is a learned instantiation of probe-first personalization-depth allocation that operates after shallow measurements are obtained.

Offline reference baselines. Under the primary GroupKFold protocol, best uniform and majority are selected using only the training fold and applied to the held-out fold (no test-oracle depth selection). Greedy needs no training. FEDPBA and context-only models are fit on the training fold only.

IV. METHOD: PROBE-FIRST ALLOCATION

Fig. 1 contrasts the offline oracle with deployable probe-first allocation. After FedAvg, each client evaluates P_0–P_2.

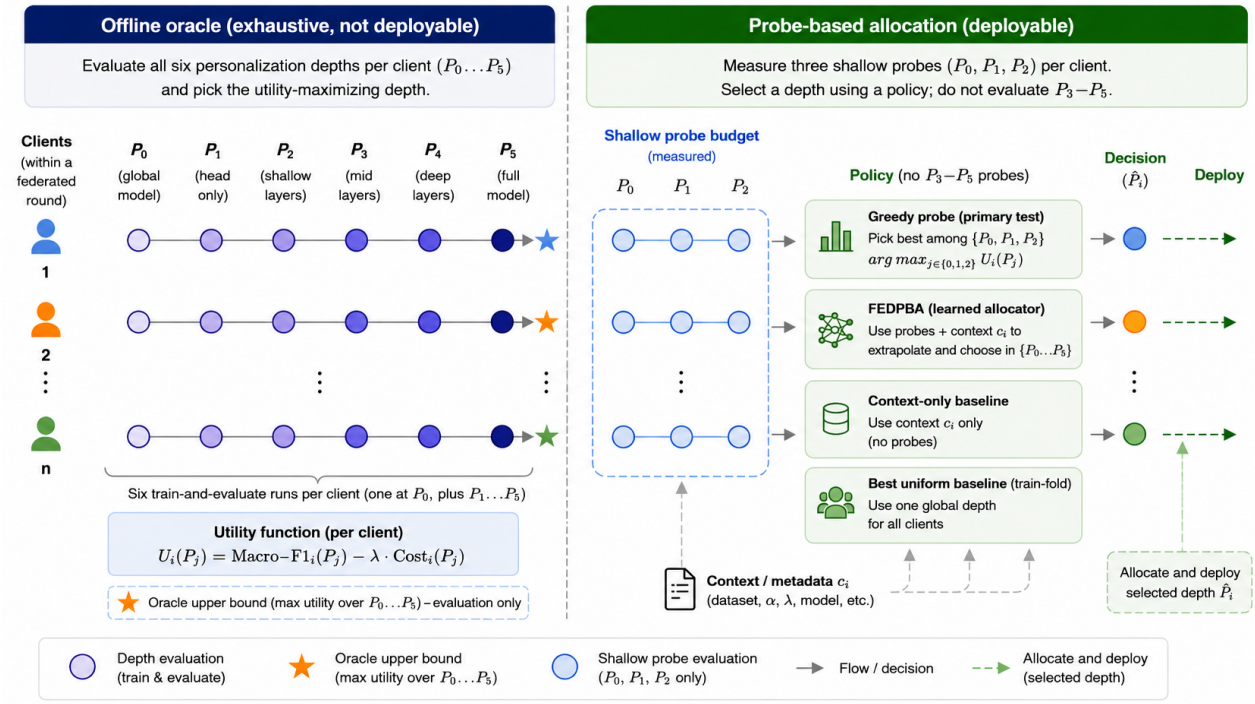


Fig. 1. Probe-first personalization-depth allocation. Left: the offline oracle exhaustively evaluates P_0–P_5 and selects the utility-maximizing depth as an empirical upper bound (not deployable). Right: three shallow probes P_0–P_2 are measured and a policy selects a depth for deployment without evaluating P_3–P_5. The training-free greedy probe rule is the direct shallow-sufficiency test; FEDPBA is the learned extrapolation policy; context-only and best uniform are information-control baselines.

The training-free greedy probe rule then stops and deploys the shallow optimum. FEDPBA is a secondary learned probe allocator: it tests whether probe signals plus context can support deep-action extrapolation to P_3–P_5. We use a histogram gradient-boosting classifier (`max_iter`=150, `max_depth`=6) because features are tabular and mixed-scale, interactions may be nonlinear, the offline policy-training set is moderate, and inference overhead is low. The use of histogram gradient boosting is not itself a methodological novelty; it serves as a controlled learned instantiation for testing whether probe information supports deep-action extrapolation. Primary evaluation of FEDPBA uses 5-fold GroupKFold by federated session: all λ copies of a client remain in the same fold; preprocessing is fit on training folds only; best uniform and majority are selected on the training fold only. Complementary leave-one-dataset-out folds stress-test *learned* transfer; because greedy is training-free, its leave-one-dataset-out aggregate equals the primary aggregate by construction and is not a cross-dataset generalization claim. Deployment never queries the oracle. Feature ablations indicate that probe outcomes outweigh representation-geometry descriptors for allocation.

V. Experimental Setup

Datasets. CIFAR-10/100 with a ResNet-style 2D encoder; PTB-XL binary Normal vs. Abnormal ECG on Lead II (WFDB) under a patient-disjoint protocol: ∼5000 records / ∼4823 patients, z-score normalized per lead and resampled to fixed length [9].

Partition (CIFAR vs. PTB-XL). N=20 clients. CIFAR samples are first partitioned across clients by label-Dirichlet $\alpha \in \{0.1, 0.5, 1.0\}$ and then split locally into 70/15/15 train/validation/test. PTB-XL follows a different patient-safe protocol: patients are first assigned to a global 70/15/15 split (once per seed; Abnormal if any record is Abnormal); then each split's patients are distributed across clients via Dirichlet α on patient-level labels. Each patient belongs to exactly one client and one global split; no second record-level split is performed inside a client. Integrity checks confirm zero global and zero cross-client patient overlap. Empty CIFAR clients are rebalanced.

Models. Encoder blocks + bottleneck adapter (pool → Linear→ReLU→Linear→ReLU) + linear head; P_1–P_5 unfreeze the blocks in Table I.

Training. FedAvg: 5 global rounds, 3 local epochs, batch 32, learning rate 10^{-3}. Personalization uses the same local epoch/LR settings on the client's train split; Macro-F1 is measured on the client's held-out test split.

Hardware / cost. Client-wide hardware multipliers scale raw cost before within-client normalization; wall time is measured during local training; communication bytes and memory are proxy estimates (parameter/activation heuristics), not hardware counters.

Lambda. $\lambda \in \{0.1, 0.3, 1.0\}$ applied post hoc on measured Macro-F1 and normalized cost.

Scale. 45 federated sessions, 900 client instances, evaluated

TABLE II
POLICY-CLASS COMPARISON (n=2700 CLIENT–λ EVALUATIONS; GROUPKFOLD).

Class	Policy	Probe evaluations	Regret	Rec. %
C	Greedy probe	3	**0.0071**	**98.3**
D	FEDPBA	3 probes + prediction	0.0110	97.3
A	Best uniform†	0‡	0.0483	88.4
B	Best context-only	0	0.0476	88.5
A	Global-only / majority	0	0.0894	78.4
A	Personalize-all	0	0.5185	−25.0

†Train-fold only selection (not online).
‡Uses full-depth outcomes for selection, not probes.

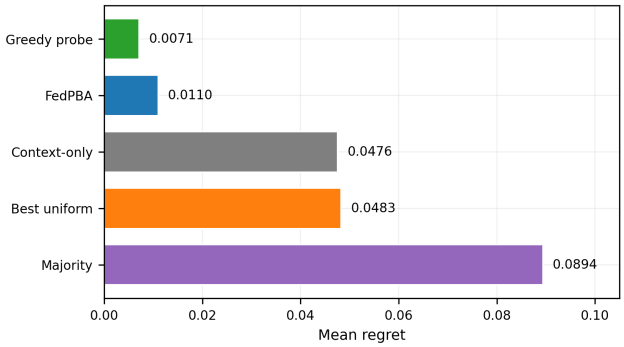


Fig. 2. Mean regret by policy class (n=2700).

under three λ values (2,700 client–λ evaluations).

Statistics. Descriptive means use the 2,700 client–λ rows. Primary inference uses n=45 session aggregates with paired tests and Holm correction; main confidence intervals are session-cluster bootstrap (B=10,000, seed 42). We do not treat n=2700 client–λ Wilcoxon tests as primary inference.

VI. RESULTS

A. Finding 1: Client-Specific Measurements Outperform Metadata-Only and Fixed Policies

As shown in Tables II–III and Fig. 2, greedy probe has the lowest regret (0.0071) and highest recovery (98.3%); FEDPBA is second (0.0110, 97.3%); best uniform (train-fold only) reaches 0.0483 (88.4%); best context-only reaches 0.0476 (88.5%). Global-only / majority reach 0.0894 / 78.4%; personalize-all yields negative aggregate utility under Eq. (1) (0.5185, −25.0%). Context-only remains approximately as weak as best uniform, whereas client-specific utility measurements produce the main reduction in regret. Relative to probe policies, context-only remains markedly worse; relative to best uniform, the difference is only marginal (0.0476 vs. 0.0483; no paired significance test).

B. Finding 2: Shallow Probes Recover Most Aggregate Oracle Utility

As shown in Table IV and Fig. 3, aggregate $\max(P_0, P_1, P_2)$ utility achieves r=0.9818 with oracle utility; individual shallow probes reach r=0.8353 (P_0), 0.8605 (P_1), and 0.8323 (P_2).

TABLE III
PROBE SUFFICIENCY METRICS.

Metric	Greedy	FEDPBA
Oracle utility recovery (%)	98.3	97.3
Probe information gain vs. context	0.0405	0.0366
Sufficiency ratio vs. uniform	0.852	0.771
Context-only recovery (%)	88.5	

TABLE IV
PROBE UTILITY VERSUS ORACLE UTILITY (PEARSON r).

Probe signal	r
P_0 utility	0.8353
P_1 utility	0.8605
P_2 utility	0.8323
$\max(P_0, P_1, P_2)$ utility	**0.9818**

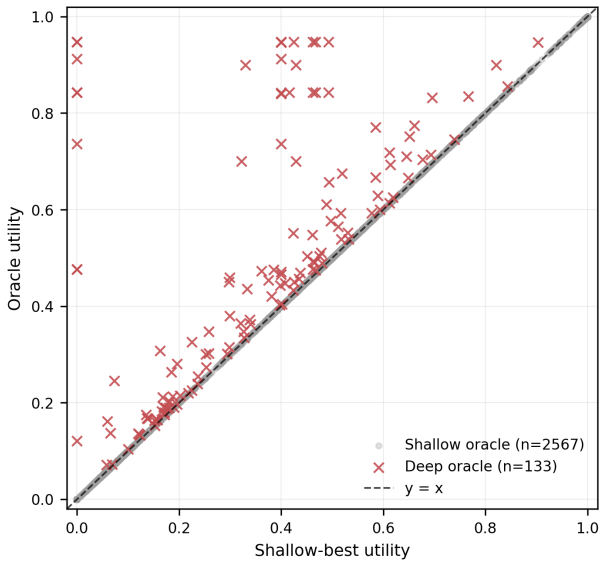


Fig. 3. Shallow-best versus oracle utility (r=0.9818 overall, n=2700; deep-oracle-only r=0.578, n=133).

On shallow-oracle rows alone, r=1 by construction. On deep-oracle rows (n=133), shallow-best vs. oracle correlation falls to r=0.578 (session-cluster 95% CI $[0.28, 0.86]$). Early shallow deltas do not positively predict deep gain in the current data (e.g., Δ_{01} vs. deep gain Pearson $r = -0.37$). The oracle selects $\{P_0, P_1, P_2\}$ in 95.1% of evaluations (2,567/2,700) and deep actions in 133 cases (4.9%). High aggregate agreement, shallow-action frequency, and deep-regime predictability should be interpreted separately. *High aggregate correlation is not equivalent to validated prediction of deep personalization gains.*

C. Finding 3: Simple Measurement Beats Allocator Complexity

The result does not show that a more complex learned allocator is required. Instead, it shows that access to client-specific shallow utility measurements is the dominant source of performance improvement: context-only ≈ best uniform, probe

TABLE V
PRIMARY PAIRED SESSION-LEVEL TESTS (n=45; RAW WILCOXON p AND HOLM-ADJUSTED p).

Comparison	$\Delta\bar{U}$	95% CI	Raw p	Holm p	d_z
Greedy vs. best unif.	+0.0411	[0.026, 0.059]	5.7×10^{-14}	1.7×10^{-13}	0.72
FEDPBA vs. best unif.	+0.0372	[0.022, 0.054]	5.7×10^{-14}	1.7×10^{-13}	0.67
Greedy vs. FEDPBA	+0.0039	[0.003, 0.005]	5.7×10^{-14}	1.7×10^{-13}	1.05

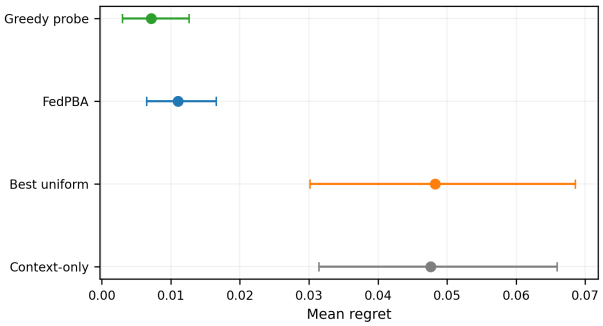


Fig. 4. Mean regret with session-cluster 95% bootstrap CI.

policies are substantially better, and the training-free greedy rule outperforms FEDPBA overall. Primary inference is session-level (n=45, Holm-corrected). Versus best uniform, greedy gains $\Delta U = +0.0411$ (d_z=0.72; raw p=5.7×10^{-14}; Holm p=1.7×10^{-13}; CI $[0.026, 0.059]$); FEDPBA gains $\Delta U = +0.0372$ (d_z=0.67; raw p=5.7×10^{-14}; Holm p=1.7×10^{-13}). Greedy still beats FEDPBA ($\Delta U = +0.0039$, d_z=1.05; Table V). As shown in Figs. 4–6, session-cluster bootstrap regret CIs are greedy $[0.0030, 0.0126]$, FEDPBA $[0.0065, 0.0166]$, and best uniform $[0.0302, 0.0686]$. Absolute means remain 0.0071 (greedy) vs. 0.0110 (FEDPBA) vs. 0.0483 (best uniform); Fig. 5 reports recovery across policies. On deep-oracle rows (n=133), mean regrets are similar (greedy 0.145, FEDPBA 0.143; session-paired utility difference CI includes zero), so FEDPBA does not consistently outperform the training-free rule even in the deep-oracle subgroup. Relative to exhaustive search, probing uses 3 vs. 6 depth evaluations (and 2 vs. 5 personalization train-and-evaluate runs, since P_0 is evaluation-only) with a median $\sim$3$\times$ wall-time speedup (Fig. 6). Leave-one-dataset-out is informative primarily for learned policies: FEDPBA regret rises from 0.0110 (grouped) to 0.0298 / 92.8% recovery, indicating calibration dependence; greedy's leave-one-dataset-out aggregate is unchanged because it has no fit parameters.

D. Finding 4: Deep-Oracle and PTB-XL Cases Define the Failure Boundary

As shown in Figs. 7–9, results vary by dataset and λ (patient-disjoint PTB-XL). CIFAR remains low-regret under probe policies (session-cluster means: CIFAR-10 greedy 0.0022, FEDPBA 0.0058, best uniform 0.0184; CIFAR-100 0.0003, 0.0014, 0.0040). Absolute regret is highest on PTB-XL (greedy 0.0189, FEDPBA 0.0259, best uniform 0.1223). By λ (Fig. 8), FEDPBA regrets are 0.0168 (λ=0.1), 0.0100 (0.3), 0.0064

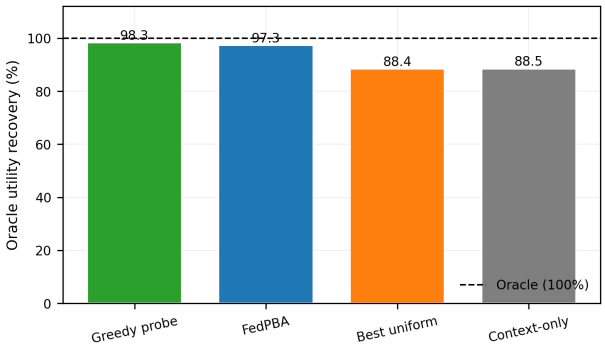


Fig. 5. Oracle utility recovery across policies (oracle mean U=0.4147).

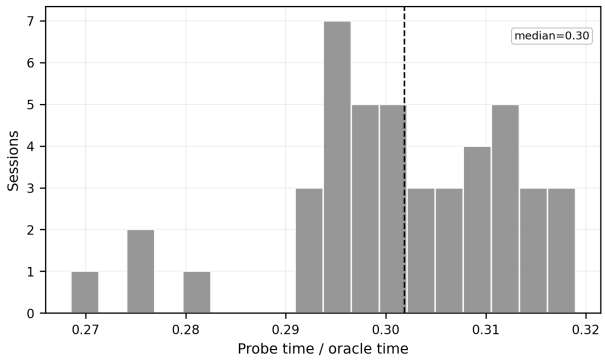


Fig. 6. Distribution of probe-to-exhaustive wall-time ratios across sessions (local personalization train and evaluation time; median probe/oracle ratio $\approx$ 0.30).

(1.0); greedy 0.0130, 0.0070, 0.0015. When the oracle prefers $\{P_3, P_4, P_5\}$, greedy has 37 $P_0\rightarrow$deep misallocations (all on PTB-XL) with mean regret 0.3665 (34 with regret $>$0.05); FEDPBA has 39 (38 PTB-XL; 1 CIFAR-10; mean regret 0.3512; 35 with regret $>$0.05) (Table VI). Across the full deep-oracle subset (n=133), mean regrets remain high for both policies (greedy 0.145, FEDPBA 0.143). These failures are not numerous in the full aggregate, but their cost is high. PTB-XL concentration aligns with known ECG site/patient heterogeneity in federated and clinical settings [26]–[28], so unconditional greedy is not a validated default for high-risk ECG deployments [25]. Recent non-IID ECG FL studies similarly report that extreme client skew and edge heterogeneity can dominate aggregate accuracy unless personalization or clustering is adapted explicitly [29], [30]. Complementary analyses further show that recording-protocol and institution shift can invalidate a single shared adaptation recipe across sites [31], while multi-institutional ECG diagnosis work emphasizes that privacy-preserving collaboration alone does not remove the need for client-conditioned decision rules when label structure differs across hospitals [32]. Deep-oracle PTB-XL cases should therefore be treated as a deployment risk regime, motivating an unvalidated but necessary escalation path when shallow probes are ambiguous.

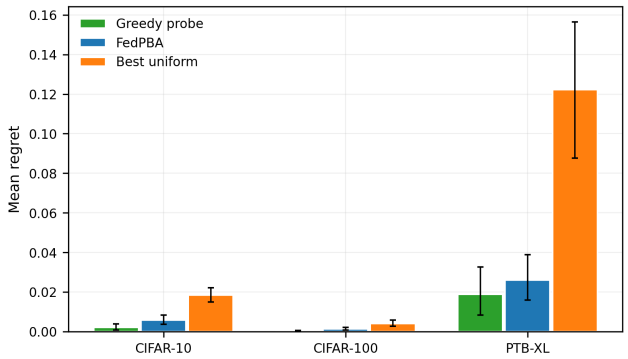


Fig. 7. Per-dataset mean regret with session-cluster 95% bootstrap CI.

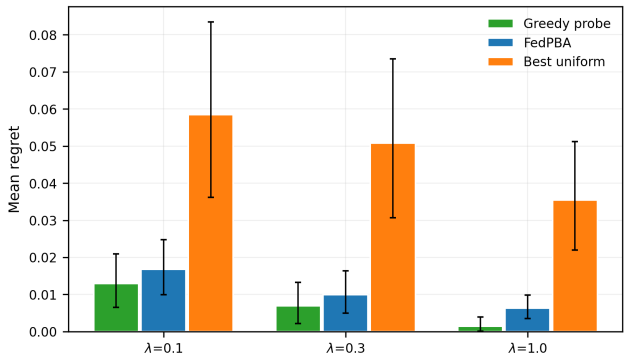


Fig. 8. Mean regret by cost sensitivity λ with session-cluster 95% bootstrap CI.

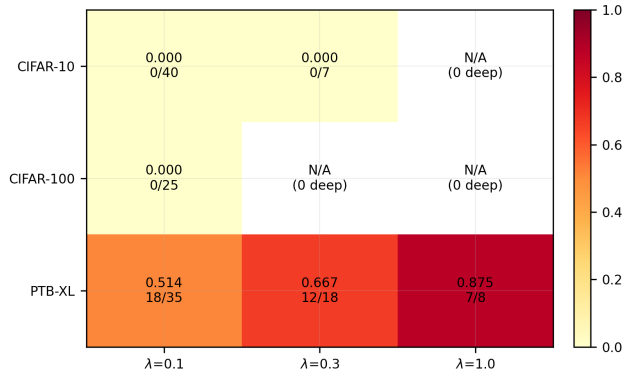


Fig. 9. Conditional rate $P(\text{greedy}{=}P_0 \mid \text{oracle deep})$ by dataset and λ (numerator/denominator over oracle-deep rows; N/A if the denominator is zero).

TABLE VI
FAILURE MODES IN WHICH P_0 IS SELECTED WHILE THE ORACLE PREFERS $\{P_3, P_4, P_5\}$.

	Greedy	FEDPBA
Count	37 (all PTB-XL)	39 (38 PTB-XL; 1 CIFAR-10)
Mean regret	0.3665	0.3512
Regret >0.05	34	35

VII. DISCUSSION

What the paper establishes. Within the evaluated regime, a limited shallow measurement budget can recover near-oracle aggregate utility: greedy probe achieves regret 0.0071 and 98.3% recovery. Probe-first allocation is therefore a promising deployment-control principle, but not a claim of universal shallow sufficiency.

Why measurement matters. Client-specific probe utilities contain information that dataset identity, α, and λ do not capture. Context-only remains close to best uniform (0.0476 vs. 0.0483) and far from probe policies, isolating measurement, not metadata, as the dominant signal.

Why learning is not the main result. The learned allocator does not outperform the training-free rule consistently. Greedy is strongest overall; on deep-oracle rows, FEDPBA does not reliably outperform it (paired session CI includes zero). Leave-one-dataset-out further increases FEDPBA regret, indicating calibration dependence. The scientific result is the value of probing, not gradient-boosting complexity. The main contribution is therefore not a claim that FEDPBA is a universally superior allocator, but evidence that probe-first decision-making is a promising control principle for cost-aware personalization.

What the high correlation does and does not mean. Aggregate $r{=}0.9818$ reflects value agreement and is partly structural on shallow-oracle rows. It does not establish that early slope predicts deep gains: deep-oracle-only correlation is $r{=}0.578$, and shallow deltas versus deep gain are weak or negative in the present data.

Failure boundary and escalation. Errors concentrate on deep-oracle rows, PTB-XL, and low λ, including high-regret $P_0{\rightarrow}$deep misallocations (Table VI). Mean regret stays elevated for both greedy and FEDPBA in the deep-oracle subset. This boundary is consistent with medical FL observations that client shift and label structure can dominate aggregate trends [26], [27]. Rising P_0–P_2 trends and probe ambiguity are candidate escalation indicators only; no gate is trained or validated here.

Relation to the companion study. The earlier study defined and characterized the offline decision problem. This study investigates an operational approximation using restricted measurements and a policy-class audit under the Probe Sufficiency Hypothesis.

Operational guidance. After FedAvg, evaluate P_0–P_2; prefer the greedy shallow optimum when λ is medium/high and domain risk is low; treat escalation indicators as unvalidated; recalibrate after architecture, population, or cost-definition changes.

VIII. LIMITATIONS AND THREATS TO VALIDITY

Findings are conditional on three datasets (vision + one binary ECG subset), ResNet-style backbones, five seeds, one P_0–P_5 ladder, a limited λ grid, and the tested cost profiles. There is no NLP, transformer, regression, or large-language-model experiment. Primary inference rests on 45 federated sessions; deep-oracle cases number only 133 client–λ rows, with failure concentration on PTB-XL. The offline oracle is an empirical upper bound, not a deployable policy. Probe-first

allocation is not cost-free: greedy still requires P_0 evaluation plus P_1–P_2 local personalization runs (three depth evaluations; two train-and-evaluate runs). There is no formal regret bound, no cryptographic privacy mechanism of the kind studied for secure real-time ECG anomaly detection [25], and no validated escalation gate. The cost model uses measured wall time together with proxy communication and memory estimates; client-wide scalar hardware multipliers cancel under within-client min–max normalization. FEDPBA depends on the calibration population (leave-one-dataset-out degrades learned transfer) and uses a standard tabular learner; its purpose is to instantiate and audit the probe-first allocation principle rather than to claim a novel classification architecture. Cost-weight and normalization sensitivity analyses support stability across the tested six$\times$two profiles, but do not exhaust all deployment cost definitions. Construct validity of "depth" is tied to our ladder. Results do not establish universal shallow sufficiency, nor do they justify immediate clinical deployment without domain-specific validation.

IX. Conclusion

This paper studies cost-aware per-client personalization-depth allocation and audits the Probe Sufficiency Hypothesis. Three shallow probes recover most aggregate oracle utility: greedy achieves regret 0.0071 and 98.3% recovery, while FEDPBA is close but not superior (0.0110 / 97.3%). High-regret deep-oracle PTB-XL failures remain, and no validated escalation detector exists. The contribution is a probe-first deployment-control principle: in this regime, client-specific measurement is more valuable than allocator complexity, but the principle remains risk-conditional.

Acknowledgments

Fig. 1 used ChatGPT image generation under IEEE AI disclosure; experiments and analyses are the authors'. Code/data pointers will be released upon acceptance.

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